Privacy-Preserving Malware Analysis with On-Prem LLMs
Reinforcement Learning Alignment for Secure SOC Automation

Introduction

- SOC overload
- AI-generated threats
- Cloud privacy risks

Problem

Cloud LLMs require sensitive telemetry → data exposure risk

Goal: Secure SOC automation

Threat Model

Adversary may observe cloud queries
Asset: telemetry

Solution: On-prem inference (no leakage)

Architecture (Figure Placeholder)

[Insert pipeline diagram here]
Reports → Distillation → SFT → PPO → GRPO

Methodology

- SFT: learn reasoning
- PPO: reward correctness
- GRPO: refine outputs

Reward Design

✓ Correctness
✓ Structured output
✓ ATT&CK terminology

Experimental Setup

- 10k malware reports
- VirusTotal, Hybrid Analysis
- LLaMA 3.2 (3B)
- H100 GPU

Results (Figure Placeholder)

[Insert charts here]
Gemini: 98%
LLaMA: 85%
Mistral: 75%

Example Output

Before: Malware detected

After:
<thought> persistence + C2 </thought>
<answer> Trojan </answer>

Conclusion

On-prem LLMs enable privacy-preserving SOC automation.

Future Work:
- Larger evaluation
- Adversarial testing
- Real-world deployment

[References & Acknowledgements here]